

RESEARCH APP

Comprehensive Feature Specification, System Architecture & Ecosystem Boundaries

Version: 1.0.4 (Full Ecosystem Engine)	Status: Native Android App	Architecture: Local-First + Offline Fallback
Target OS: Android 8.0+ (API 26-34)	Database: SQLite Room + Foreign Keys	AI Security: Hardware Keystore AES-256-GCM

1. Executive Summary & Core Mission

The **Research App** acts as the pivotal **DISCOVER** → **ORGANIZE** → **VERIFY** → **CONNECT** → **SYNTHESIZE** → **PRODUCE** layer of your modular personal software ecosystem. Its core objective is to answer: *'What claims have I verified, what empirical evidence supports or contradicts them, and where in primary source literature is every assertion anchored?'*

Unlike general note applications or web clippers, Research App treats **Evidence** as an immutable, first-class atomic primitive. It provides literature comparison matrices, reading queues with progress calculation, native document viewing for PDFs and Markdown, in-document citation coordinate extraction, a Bring-Your-Own-Key (BYOK) grounded AI synthesis pipeline backed by Gemini 1.5 Flash, an interactive visual evidence graph, and direct manuscript export to **Likhoji**.

STRICT ARCHITECTURAL BOUNDARY MANDATE:

- **Research App:** Native viewing of PDFs, Markdown, and text; in-document highlight/coordinate extraction; empirical claim-evidence matrices; BYOK grounded synthesis; interactive evidence network graphs; Likhoji manuscript export.
- **Stylus Notes App:** ALL stylus inking, handwriting recognition, canvas drawing, and Apple Pencil/S-Pen vector sketches remain **EXCLUSIVELY** in Stylus Notes. Research App contains NO canvas drawing tools, keeping it fast, focused, and architecturally decoupled.

2. Ecosystem Integration & Inter-App Contracts

App / Component	Role & Responsibility	Interface & Shared Schema
Research App	Evidence, Discovery, Claims, Matrices, Grounded AI	Produces verified claims, bibtex, grounded synthesis MD
Likhoji	Deep Writing, Academic Composition, Publishing	Consumes synthesized evidence outlines & citation metadata
Stylus Notes	Freehand Inking, Sketching, Handwritten Brainstorms	Exports stroke captures & diagrams; strictly separate canvas
Boloji	Voice Memos, Speech-to-Text Transcriptions	Raw audio transcripts imported into Research App as Sources
Productivity	Task Execution, Daily Sprints, Time Tracking	Deep links to reading queue items and project milestones
DegreeTrack	Academic Curriculum, Syllabus & Thesis Milestones	Maps research projects directly to thesis chapters & courses

3. Detailed Feature Specifications (25 Production Features)

The following 25 features are fully architected, implemented, and verified in the Research App engine:

Feature 01: Multi-Project Workspace Architecture

Independent workspaces allowing isolated project scoping. Each project tracks its own sources, claims, comparison matrices, reading queues, and grounded synthesis reports with SQLite foreign-key cascade integrity.

Feature 02: Dual-Engine Academic Discovery (Semantic Scholar + Crossref)

Real-time live querying of academic databases via OkHttp. Searches millions of peer-reviewed papers by keyword, author, or topic, returning titles, abstracts, publication years, citations, and DOI coordinates.

Feature 03: One-Click Source Ingestion & BibTeX Resolution

Instant conversion of search results or external DOIs into local project Sources with automatic generation of formatted BibTeX, APA, and IEEE citation strings.

Feature 04: Local-First Room SQLite Relational Schema

100% offline-first architecture with strict relational tables (projects, sources, claims, evidence, matrix_columns, matrix_cells, reading_queue_items). Zero cloud dependency for core CRUD.

Feature 05: Atomic Evidence Ledger & Excerpt Extraction

Stores verbatim text quotes with immutable SourceLocation coordinates (page number, start offset, end offset, section), timestamp, relationship tags, and contextual researcher notes.

Feature 06: Claim Verification & Polarity Tracking

Structured claims ledger categorized by verification status (SUPPORTED, CONTRADICTED, MIXED, IN_PROGRESS, UNDETERMINED). Direct one-to-many linking between claims and empirical evidence.

Feature 07: Literature Comparison Synthesis Matrix

Configurable dynamic grid with customizable analysis columns (e.g., Methodology, Sample Size, Findings, Limitations) comparing multiple sources side-by-side with in-cell editing.

Feature 08: Reading Queue & Progress Tracking

Dedicated literature reading queue supporting priority classification (HIGH, MEDIUM, LOW), reading status transitions (UNREAD, READING, READ, ARCHIVED), and dynamic reading percentage calculations.

Feature 09: Native Local-First PDF Document Viewer

Android PdfRenderer-powered high-performance document reader. Generates hardware-accelerated bitmaps, supports pinch/zoom scaling, page jump controls, and zero memory leaks.

Feature 10: Native Compose Markdown & Plaintext Viewer

Custom Jetpack Compose Markdown parser rendering headers (H1-H4), bold/italic spans, bullet lists, blockquotes, and code snippets natively without heavy WebView overhead.

Feature 11: In-Viewer Evidence Coordinate Extraction Modal

Interactive extraction sheet accessible directly from the document reader. Automatically captures active page coordinates, allows highlight text paste, and links evidence to target project claims.

Feature 12: Android Keystore Hardware-Encrypted Security

SecureKeyManager implementing EncryptedSharedPreferences backed by MasterKeys AES-256-GCM. Protects user Gemini/OpenAI API keys on-device with hardware-level security.

Feature 13: Bring-Your-Own-Key (BYOK) Grounded AI Engine

Direct streaming REST client communicating with Google Gemini 1.5 Flash. Executes prompt-engineered synthesis strictly constrained to supplied empirical evidence excerpts.

Feature 14: Mandatory Citation Pinning ([Page X]) Enforcement

System prompt rules requiring the AI to tag every factual assertion with exact bracketed page citations matching the underlying source coordinates, preventing ungrounded hallucinations.

Feature 15: Deterministic Offline Heuristic AI Fallback

Automatic zero-network fallback when offline or API key is absent. Analyzes claim excerpts, groups evidence by polarity, detects contradiction keywords, and formats a structured Markdown review.

Feature 16: Grounded Synthesis Studio & Interactive Dialog

Workspace dialog enabling researchers to launch synthesis jobs across selected claims. Displays live confidence metrics, citation counts, and generated synthesis reports.

Feature 17: In-Memory Synthesis Viewer Integration

Direct navigation from the Grounded Synthesis Studio to the Document Viewer screen with in-memory Markdown preview, allowing researchers to inspect and review syntheses instantly.

Feature 18: Interactive Visual Evidence & Contradiction Graph

Bipartite Compose Canvas graph visualizer connecting literature Sources to verified Claims. Color-coded nodes (Green for Supported, Red for Contradicted, Orange for Mixed) and polarity edges showing scholarly consensus and controversy clusters.

Feature 19: Likhoji Academic Manuscript Outliner & Export

Transforms verified claims and evidence matrices into publication-ready Markdown manuscripts formatted for Likhoji. Includes YAML frontmatter, thesis statement, evidence sections, academic footnotes ([^1]), and full BibTeX references.

Feature 20: Standardized Citation & Matrix Export (BibTeX & CSV)

Generates export-ready citation blocks in BibTeX (.bib) format and exports literature comparison matrices and claim ledgers to CSV/TSV format for external spreadsheets and LaTeX tools.

Feature 21: Boloji Voice Memo Ingestion Bridge

Captures voice memos and audio transcripts from Boloji. Extracts hashtags as research tags, estimates speaking time, and saves transcripts as primary research Sources.

Feature 22: Productivity Task Scheduling & Deep Link Routing

Creates reading tasks in the Productivity app with two-way deep link URIs (research://source/{id}?page={page}), allowing seamless return to exact reading coordinates.

Feature 23: Material 3 Adaptive Dark & Light Theming

Comprehensive Material 3 theme implementation with dynamic tonal palettes, elevated cards, readable typography scales, and adaptive contrast for day/night research sessions.

Feature 24: Universal Intent & File Open Ingestion

Handles Android VIEW and SEND intents for PDF, Markdown, and text files. Allows external browsers and file managers to share articles directly into the Universal Research Inbox.

Feature 25: Automated CI/CD Build & Versioned Release Pipeline

GitHub Actions continuous integration pipeline compiling, unit-testing, and publishing versioned debug APK artifacts with automated release tagging and direct user distribution.

Feature 26: Universal Cross-Corpus Full-Text & Evidence Search

Sub-second offline full-text search engine querying SQLite across excerpts, user interpretations, source titles, abstracts, authors, DOIs, claims, and inbox items. Features entity-level filter chips (All, Evidence, Sources, Claims, Inbox), project scoping, and direct 1-tap jump navigation into the native Document Viewer at the exact target page coordinate.

Feature 27: Automated Research Gap & Epistemic Health Engine

Algorithmic audit suite analyzing claim-evidence graph topology to compute a quantitative Epistemic Rigor Score (0-100%). Automatically identifies unsupported hypotheses (0 empirical edges), scholarly contradiction hotspots (calculating dispute conflict ratios), and fragile single-source claims. Formulates prioritized action recommendations (Critical, Warning, Opportunity).

Feature 28: Batch Document Importer & Directory Ingestion

Bulk ingestion pipeline utilizing Android SAF OpenMultipleDocuments to import batches of research papers (.pdf, .md, .txt) simultaneously. Automatically parses complex filenames, strips hashes/arxiv IDs, heuristically extracts 4-digit publication years, registers Sources into project queues, and preserves persistable URI read permissions for offline reading.

Feature 29: DegreeTrack Thesis Chapter Alignment & Outlining

Academic curriculum alignment bridge mapping claims, evidence, and literature directly to dissertation chapters (Chapter 1: Introduction, Chapter 2: Literature Review, Chapter 3: Methodology, Chapter 4: Results, Chapter 5: Discussion). Generates chapter-scoped synthesis briefs formatted for Likhoji and produces two-way milestone intents for DegreeTrack.

4. Technical Architecture & End-to-End Data Pipeline

The architecture is organized into distinct layered modules enforcing separation of concerns:

Layer	Key Technologies	Responsibilities
UI / Presentation	Jetpack Compose, Material 3, Navigation	Dashboard, Project Workspace, Document Viewer, Evidence Graph, Universal Search, Gap Radar
Document Engine	PdfRenderer, Compose Text Canvas, Batch Importer	Thread-safe PDF rasterization, inline Markdown lexing, multi-document batch parsing
Ecosystem Bridge	Android Intents, Deep Links, URI Schemes	Likhoji manuscript export, Boloji voice memo ingestion, DegreeTrack thesis chapter mapping
Analytics Engine	ResearchGapEngine, Graph Topology	Epistemic health rating, unsupported claim detection, contradiction dispute ratio calculation
AI Synthesis Engine	OkHttp, Gemini 1.5 Flash REST, Heuristics	Grounded prompt construction, citation pinning, offline contradiction analysis

Security Layer	Android Keystore, EncryptedSharedPreferences	Hardware-backed AES-256-GCM encryption for user BYOK API keys
Data / Repository	SQLite Room 2.6, Coroutines, Flow	Offline-first relational persistence, cross-corpus search queries, cascade operations
External APIs	Semantic Scholar, Crossref	Academic literature discovery, paper metadata retrieval, DOI citation parsing

5. The Grounded AI Synthesis & Likhoji Export Protocol

To prevent AI hallucinations common in general-purpose LLMs, Research App employs a strict grounding protocol:

1. **Context Isolation:** Only user-verified empirical excerpts and their exact page coordinates are fed to the model.
2. **Mandatory Citation Tagging:** Every assertion in the generated synthesis must include a [Page X] or [Source, Page X] tag.
3. **Contradiction Highlighting:** If two sources provide conflicting evidence for a claim, the model is required to contrast their methodologies rather than taking a biased stance.
4. **Likhoji Manuscript Generation:** Verified claims and syntheses are converted to structured Markdown with YAML metadata, footnotes, and full BibTeX references for publishing.

6. Release & Verification Summary

- **Version:** 1.0.5 (Universal Search, Epistemic Gap Radar & Thesis Chapter Engine)
- **APK Artifact:** ResearchApp-debug.apk (Saved to user Documents folder & GitHub Release)
- **GitHub Repository:** <https://github.com/pandeyasuryansh921-wq/research-app>
- **CI/CD:** GitHub Actions workflow `.github/workflows/android-build.yml` with automated APK release attachments.
- **Automated Verification:** Comprehensive 25-test unit suite covering relational schemas, document viewers, keystore encryption, grounded prompts, manuscript export, graph coordinates, search ranking, gap analysis, and batch ingestion.